



DPUSGEDistortionBezierParamsExt

Allows modification of the geometry correction provided by DPUSGEDistortionBezier5Ext while the simulation is running.

1. Dynamic parameters:

Zoom Defines the width of the DPU's local coordinate system, which is used to position the grids with the ViewX/Y/W/H parameters.
Default: 3250

2. Static parameters:

WindowX, WindowY Position of the window, in screen coordinates (pixels).
Default: (0, 0) (the top-left edge of the screen)

WindowW, WindowH Size of the window, in pixels.
Default: (1024, 768)

Views Number of views that are being configured (1..10)

ViewX[01..10], ViewY[01..10] Position of the grid for each displayed view, in the DPU's local coordinate system

ViewW[01..10], ViewH[01..10] Size of the grid for each displayed view, in the DPU's local coordinate system

ViewServer[01..10] Name of the computer that renders each displayed view.

ViewPort[01..10] TCP port number of the server DPU for the displayed view. The default value only needs to be changed if more than one DPUSGEDistortionBezier5Ext is running on the same computer.

3. Dependencies to other DPUs:

DPUSGEDistortionBezier5Ext must run on the remote computers specified under *ViewServer[10]*.

DPUSGEDistortionBezier5ExtSaveParams should be used to save the modified geometry correction after a restart of SILAB.

4. Functional principle:

Allows modification of the geometry correction provided by DPUSGEDistortionBezier5Ext while the simulation is running. A graphical window displays the correction grids of all connected distortion DPU instances. Using mouse and keyboard controls, individual control points, or groups of points, can be selected and moved. Adjustments can be made in coarse or fine precision, so that good correction can be achieved quickly.

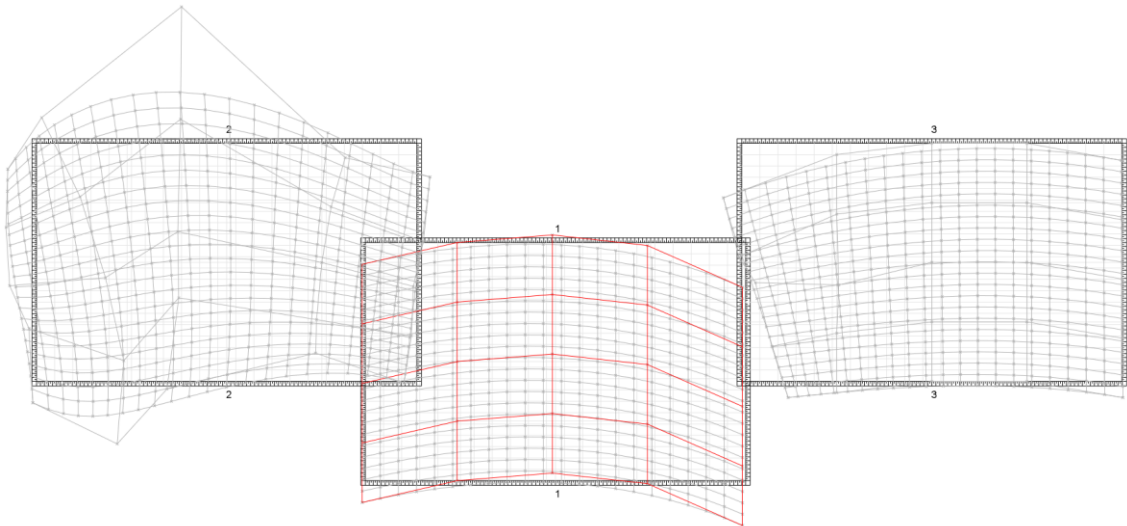


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An example of the window is seen in the following screenshot.

View = 1 Kontrollgitter
[F1] : Hilfestellung



5. Keyboard and Mouse Controls:

While the simulation is running, the following keyboard and mouse commands are used to modify the geometry correction:

[F1] Help on keyboard and mouse controls

Switching views:

[1]...[0]	Selecting the current view
+ [SHIFT]	Center current view
[G]	Switch to Grid points
[C]	Switch to Control points
(MMB)	Zoom in/out
[PG-Up/Down]	Zoom in/out
(RMB)	Move the view
[ALT]+[CUR]	Move the view

Selecting and moving points:

(LMB)	Select a point
(LMB+SHIFT)	Extend the selection
(LMB+Y)	Select a column
(LMB+X)	Select a row
[CUR]	move points quickly

together with:

[CONTROL]	half speed
[SHIFT]	quarter speed
[CONTROL+SHIFT]	1/20th speed
[A]+[SHIFT]	Select all points



[CONTROL]+[ALT]+[R] Reset points to reference grid

Show grid in the views:

[F5]	Disable grid display
[F6]	Show control points
[F7]	Show grid points
[F8]	Show controls and fine grid
[ESC]	Hide help message

6. Example:

This DPU works well using the following script:

```
# Window to configure distortion, displayed on the operator
DPUSGEdistortionBezierParamsExt SGEdistParams
{
    Computer = {OPERATOR};
    Index = 0;
    ViewServer01 = "M2VISF1";

    ViewW01 = 1920;
    ViewH01 = 1200;

    ViewX01 = -960;
    ViewY01 = 0;

    WindowX = 0;
    WindowW = 1920;
    WindowH = 1024;
    Views = 1;
};

# DPU saving distortion parameters, running on the operator
DPUSGEdistortionBezier5ExtSaveParams SaveParamsFront
{
    Computer = {OPERATOR};
    Index = 20;
    Server = "M2VISF1";
    DistInstanceName = "M2VISF1.SGEdist.";
    FileName = "DistFront";
    SaveDateTime = 5000;
};

# DPU for actual distortion, running on the visualization machines
DPUSGEdistortionBezier5Ext SGEdist
{
```



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```
Computer = {VISF};
Index = 40;
ResolutionX = 1920;
ResolutionY = 1200;
SubdivX = 21;
SubdivY = 21;
# Use an equal distribution of control points initially
Points = (
    0, 0,      480, 0,    960, 0,    1440, 0,    1920, 0,
    0, 300,    480, 300,  960, 300,  1440, 300,  1920, 300,
    0, 600,    480, 600,  960, 600,  1440, 600,  1920, 600,
    0, 900,    480, 900,  960, 900,  1440, 900,  1920, 900,
    0, 1200,   480, 1200, 960, 1200, 1440, 1200, 1920, 1200
);

};

# Load the configuration files saved by the SaveParamsFront instance above
#includeif system/DistFront.inc
```